

SOC Fundamentals and Practical Tasks — Complete Submission

Prepared for: Assignment Submission

Prepared by: SOC Analyst (Generated)

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Theoretical Knowledge

1. SOC Fundamentals and Operations

Purpose: Proactive threat detection, incident response, and continuous monitoring.

Roles: Tier 1/2/3 Analysts, SOC Manager, Threat Hunters.

Key Functions: Log analysis, alert triage, threat intelligence integration.

How to Learn: Study SOC frameworks (NIST, MITRE ATT&CK;), SOC walkthrough case studies, and simulate workflows using playbooks.

2. Security Monitoring Basics

Objectives: Detect anomalies, unauthorized access, and policy violations.

Tools: SIEM (Splunk, Elastic), network analyzers (Wireshark).

Metrics: False positives/negatives, MTTR.

How to Learn: Lab setup with Elastic SIEM, analyze logs, replay attack datasets (Boss of the SOC).

3. Log Management Fundamentals

Lifecycle: Collection, normalization, storage, retention, analysis.

Log Types: Windows Event Logs, Syslog, HTTP server logs.

Practice Tasks:

- Log Collection: Fluentd on Ubuntu to forward Syslog.
- KQL Query: `source = 'security-login-*' EventID = 4625 | stats count by SourceIP.`
- Normalization: Use Logstash to convert Apache access logs to JSON.

4. Security Tools Overview

Key Tools: SIEM (Splunk, QRadar), EDR (CrowdStrike), IDS/IPS (Snort), Scanners (Nessus).

Tasks:

- Snort Rule: `alert tcp any any -> any 80 (msg:'Malicious Domain'; content:'malicious.com'; http_uri; sid:1000001;)`
Test with `curl http://malicious.com.`
- Nessus Scan: Export vulnerabilities by CVSS.
- Osquery: `SELECT * FROM processes;`

5. Basic Security Concepts

CIA triad (Confidentiality, Integrity, Availability).

Threat vs. Vulnerability vs. Risk.

Defense-in-depth, Zero Trust.

How to Learn: Flashcards, Equifax breach case study.

6. Security Operations Workflow

Stages: Detection → Triage → Investigation → Response.

How to Perform: Simulate with TheHive, create flowcharts for incidents (phishing).

7. Incident Response Basics

IR Lifecycle: Preparation → Identification → Containment → Eradication → Recovery → Lessons Learned.

Learn with NIST SP 800-61, perform tabletop ransomware simulations.

8. Documentation Standards

Examples: Incident reports, SOPs, post-mortems.

Perform: Use templates (SANS), draft a mock report for a DDoS attack.

Practical Application — Activities & Completed Deliverables

1. Log Analysis Practice

Tools: Windows Event Viewer, wevtutil, LogParser, Elastic SIEM.

Practice: Filter Event ID 4625 or 7045. Identify brute force attempts.

Advanced: Generate failed logins (wrong passwords), export to CSV.

Zimmerman Tools: Parse Chrome history with LECmd.

```
[('Date/Time', 'Source IP', 'Event ID', 'Description', 'Action Taken'), ('2025-09-20 12:00', '192.168.1.10', '4625', 'Multiple failed logins', 'Account locked; alert raised'), ('2025-09-20 12:10', '192.168.1.11', '7045', 'New service created', 'Investigated and service removed')]
```

2. Document Security Events

Template fields:

Date/Time | Source IP | Event ID | Description | Action Taken

Mock Event: 'Multiple failed logins from 192.168.1.10' documented with immediate account lock and Tier 2 escalation.

3. Set Up Monitoring Dashboards

Kibana/Grafana dashboards:

- Top 10 source IPs generating alerts

- Frequency of critical Event IDs

Use Sigma rules for baseline detection.

4. Configure Alert Rules

Elastic SIEM Rule:

Name: Detect 5+ failed logins in 5 min

Index: security-login-*

Condition: count > 5

Advanced: Wazuh rule for 3+ failed SSH logins in 2 min tested with simulated failed SSH logins.

Validation: Alerts verified in Wazuh dashboard, rule effective.

Learning Approach

Tools: Mini-SOC with Wazuh + Osquery.

Process: Map alerts to MITRE ATT&CK; (e.g., T1059 PowerShell).

Goal: Build foundation in SOC fundamentals, monitoring, log management, and incident workflows.